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Keywords

3d vision, depth estimation, foundation models, signal processing and analysis, survey

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Monocular Depth Estimation in the Foundation Model Era: A Survey

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Abstract—Monocular Depth Estimation (MDE) is one of the most fundamental tasks in 3D computer vision, crucial for spatial understanding, 3D reconstruction, robotics, and autonomous driving. In recent years, the field has undergone a paradigm shift from single-domain training toward multi-domain learning using large-scale datasets. Models developed under this paradigm are commonly referred to as foundation models, and they have demonstrated remarkable potential for real-world generalization. While foundation models have rapidly become a dominant approach to MDE, their strengths and limitations remain insufficiently understood. In this article, we present the first comprehensive review of foundation models for MDE, introducing a novel taxonomy that organizes existing methods by their problem-driven objectives and core algorithmic contributions, enabling a structured understanding of technological advances in the field. Additionally, we investigate prevalent model adaptation strategies employed to enhance depth prediction in diverse real-world scenarios. We further conduct an extensive performance evaluation on benchmark datasets, offering quantitative comparisons across representative approaches. Finally, we outline open challenges and promising future directions to guide research towards more robust, versatile, and efficient MDE foundation models.

Index Terms—Depth Estimation, Foundation Models, 3D Vision, Survey

1 Introduction

Depth information serves as a cornerstone for a broad spectrum of applications, such as 3D reconstruction [1], robotic perception and navigation [2], and autonomous driving [3]. Traditionally, active sensing devices, e.g., LiDAR, Time-of-Flight sensors, and ultrasonic probes offer high precision. However, their adoption in real-world scenarios is hindered by high costs, considerable processing time, operational complexity, low spatial resolution, and susceptibility to noise [4], [5]. Driven by advancements in artificial intelligence, learning based Monocular Depth Estimation (MDE) has emerged as a promising alternative. MDE estimates depth from single monocular images or video sequences, providing a cost-effective solution with an unrestricted depth range, high-resolution output, and ease of integration across diverse robotic platforms.

Earlier approaches to MDE have primarily relied on a single-domain training and testing paradigm, where a variety of network architectures and training strategies have been proposed. To achieve accurate, high-resolution depth maps, prior works explored encoder–decoder networks [6], dilated convolutional networks [7], multi-scale frameworks [8], and vision transformers [9]. Building on attention mechanisms, several methods [10], [11] employed global or temporal attention to strengthen feature integration, while recent successes in diffusion models inspired conditional denoising frameworks [12], [13] for depth estimation. However, these methods are typically restricted to fixed domains and therefore exhibit limited generalization, which significantly constrains their applicability in practical, real-world settings.

With the validation and emergence of scaling laws in text-to-image generation and video generation models [14], the concept of foundation models has been formally established. These models are trained on broad datasets. Once trained, they exhibit emergent zero-shot generalization capabilities, which allow them to be adapted to diverse downstream tasks [15]. Pioneering works such as MiDaS [16] and Metric3D [17] marked the beginning of Monocular Depth Foundation Models (MDFMs). Since then, the field has advanced at an exceptionally rapid pace [18]–[21], with notable improvements driven by scaling up the volume and diversity of training data, adopting high-capacity network architectures from other computer vision domains, and refining supervised strategies.

However, the sheer speed of progress and the breadth of emerging methods make it increasingly challenging for researchers to stay abreast of the latest developments and to fully grasp the underlying technical details. Although several survey papers on depth estimation are available, they all [22]–[27] focus on early techniques developed prior to the advent of foundation models. Only two existing surveys mention depth foundation models. Zhang et al. [28] briefly list key frameworks from previous depth foundation models but cover fewer than 15 monocular methods. Xu et al. [29] focus on the evolution toward foundation models in deep learning-based depth estimation, reviewing works across monocular image, stereo image, multi-view image, and monocular video settings; however, only 19 of the studies they review specifically focus on MDFMs.

Given this context, despite the rapid advancements in MDFMs, to the best of our knowledge, no comprehensive and systematic investigation focusing exclusively on

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MDFMs has yet been conducted. This absence creates significant challenges for the community in tracking the latest developments and identifying promising future directions. Motivated by this gap, we present a timely and exhaustive review of recent progress in foundation models for MDE. Our work delivers a structured and in-depth analysis of over 90 papers published or under peer review up to October 2025. Crucially, we introduce, for the first time, a taxonomy that classifies existing methods through a joint analysis of their problem-driven objectives and core algorithmic contributions, thereby providing clear guidance for researchers and practitioners navigating this rapidly evolving field. Additionally, we examine existing strategies for adapting foundation models for real-world applications and compare their performance across diverse depth estimation benchmarks, enabling a fair and comprehensive assessment of their practical utility.

Our principal contributions are summarized as follows:

- We establish the urgent need for a dedicated and comprehensive survey on Monocular Depth Foundation Models (MDFMs), motivated by rapid advancements, the difficulty of tracking recent progress, and the complexity of emerging technical details.
- We propose a novel, structured taxonomy that organizes existing methods by their problem-driven objectives and core algorithmic contributions, providing the community with a clear framework for understanding current progress and guiding future research directions.
- We conduct a systematic analysis of prevalent adaptation strategies for tailoring MDFMs to diverse real-world domains, elucidating their principles, strengths, and limitations, and providing insights for practical deployment.
- We provide extensive experimental comparisons and analyses to enable a fair, standardized, and comprehensive evaluation of MDFM capabilities and utility across varied application scenarios.
- We present open challenges and offer informed perspectives to guide valuable future researches.

The remainder of this paper is structured as follows. Sec. 2 provides background knowledge of MDE and foundation models, and introduces a systematic taxonomy of existing solutions. Sec. 3-6 present an in-depth analysis of the key contributions and technical advances in prior works within the scope of our taxonomy. Sec. 7 examines prevalent adaptation strategies for real-world deployment, while Sec. 8 offers a comprehensive evaluation of existing MDFMs from multiple perspectives. Sec. 9 highlights promising directions and Sec. 10 summarizes our paper.

2 Relevant Definition and Taxonomy

2.1 Problem Definition

MDE is the task that infers per-pixel depth from a single RGB image. Formally, let $I \in \mathbb{R}^{3\times H\times W}$ represent a RGB image, where H and W are the height and width, respectively. MDE estimates depth map $D \in \mathbb{R}^{H\times W}$ from I, defined as:

$$D = f_\theta(I), \tag{1}$$

where $f_\theta(\cdot)$ denotes the depth network with learnable parameters θ. Monocular video depth estimation extends MDE to sequential inputs, predicting temporally coherent depth maps from sequential frames. Let $\mathbf{I} = \{I_t \in \mathbb{R}^{3\times H\times W} \mid t = 1, 2, \dots, T\}$ represent a sequence captured by a monocular camera. At time step t, the network leverages multiple consecutive frames to estimate the current depth map:

$$D_t = f_\theta(I_{t-k}, \dots, I_t, \dots, I_{t+k}), \tag{2}$$

where $k \geq 0$ defines the temporal range for prediction, D_t is the estimated depth map of frame I_t. When $k = 0$, Eq. (2) transforms into the single-image formulation in Eq. (1).

The defining property of depth foundation models is *robust generalization*: the ability to maintain consistency across variations in environment, camera configuration, and visual domain. The network $f_\theta(\cdot)$ in the foundation model for MDE typically employs complex, large-scale architectures with a large number of parameters, e.g., Vision Transformers (ViTs) [10] and diffusion models [12], [13]. These models are trained on mixed datasets with sophisticated and meticulously designed learning strategies. However, this capability is intrinsically connected to four fundamental challenges identified in our taxonomy:

1) Expanding training horizons: overcoming data scarcity across diverse domains through pseudo-label generation and synthetic-data-driven augmentation.
2) Preserving geometric fidelity: maintaining sharp boundaries and fine-scale structural details in predicted depth maps.
3) Resolving scale ambiguity: addressing inconsistencies introduced by variations in camera intrinsics, field of view, and lens distortion.
4) Ensuring temporal consistency: achieving stable, coherent depth predictions across sequential frames in monocular video.

By situating the definition of foundation models within these four challenges, we establish a direct connection between the core model properties and the primary axes of innovation in MDE.

2.2 Taxonomy

Fig. 1 presents our taxonomy of MDFMs, organized according to the technical strategies used to address four fundamental challenges in the field. The taxonomy captures the progressive broadening of scope, from expanding training data diversity to preserving fine-grained structures, improving pixel–depth mapping fidelity, and ensuring temporal stability in sequential settings.

2.2.1 Expanding Training Horizons: Multi-Source Data Extension

Motivation: MDFMs rely on large-scale, high-quality, and diverse datasets. However, depth annotation pipelines such as active sensors, stereo matching, and Structure-from-Motion (SfM) are difficult to deploy in niche environments (e.g., underwater, aerial) and often produce sparse or noisy labels, impairing the training quality.

Key Strategies: Two dominant approaches expand data coverage: 1) *Pseudo-label generation in semi-supervised learning*

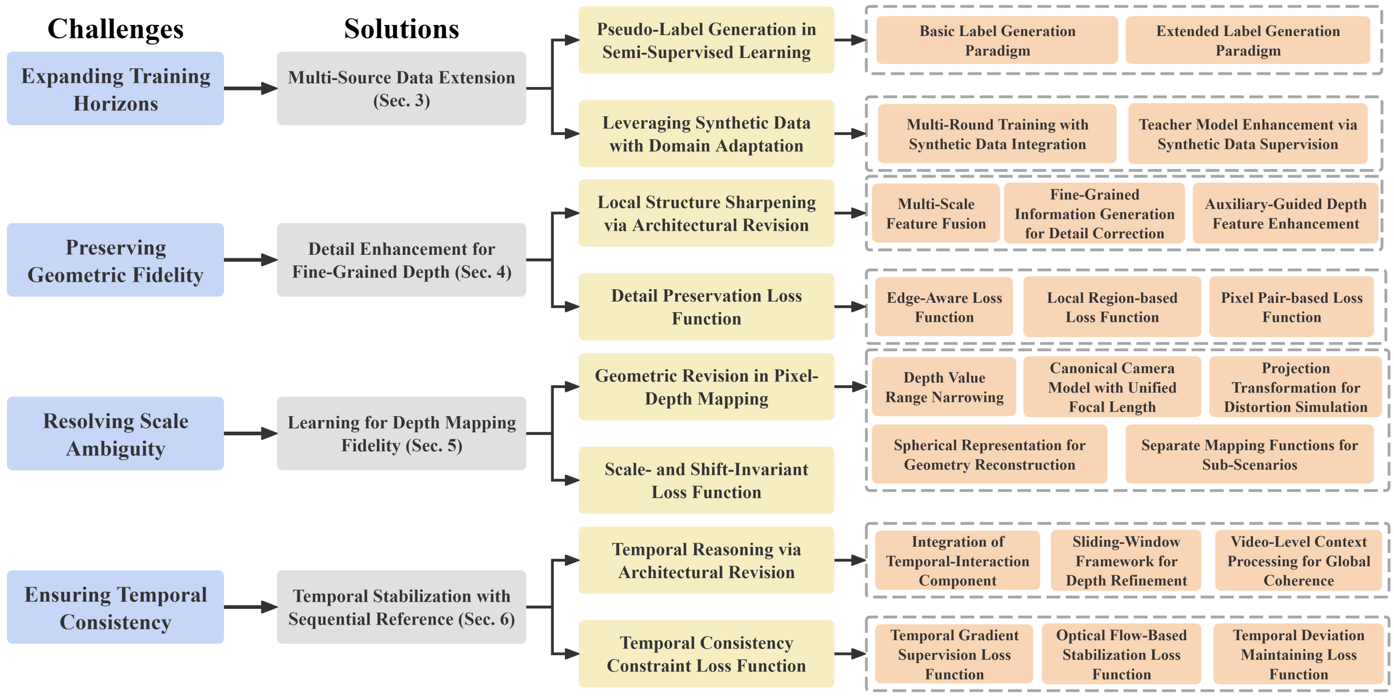

Fig. 1. Diagram of the proposed taxonomy, derived from a joint analysis of technical contributions and the essential challenges they target.

using pre-trained depth models to generate dense pseudo labels for unlabeled real images, with methods either applying direct pseudo-label generation or refining labels via confidence filtering and domain-specific adjustments. 2) *Leveraging synthetic data with domain adaptation* exploiting pixel-perfect labels in synthetic while mitigating domain gaps via multi-round training or synthetic-data-supervised teacher models.

These techniques substantially enlarge training diversity, enhance training quality, and reduce annotation cost.

2.2.2 Preserving Geometric Fidelity: Detail Enhancement for Fine-Grained Depth

Motivation: While MDFMs often ensure global depth consistency, they may oversmooth local details, causing blurred boundaries and loss of fine structures, which are critical in applications requiring precise geometry (e.g., robotics).
Key Strategies: 1) *Local structure sharpening via architectural revision* improving local detail retention via multi-scale feature fusion, complementary high-resolution depth branches, or auxiliary-guided refinement (e.g., semantic cues). 2) *Detail-preservation loss function* prioritizing boundaries and local structures during optimization using edge-aware, local-region-based, or pixel-pair-based objectives.

Both strategies counteract detail erosion, enabling sharper and more accurate depth maps.

2.2.3 Resolving Scale Ambiguity: Learning for Depth Mapping Fidelity

Motivation: Variations in camera parameters (focal length, field of view, lens distortion) alter pixel–depth relationships, introducing scale ambiguity and hindering cross-domain transfer.
Key Strategies: 1) *Geometric revision in pixel-depth mapping* standardizing depth mapping across scenarios using techniques such as depth-range normalization, canonical camera models, projection transformations for distortion simulation, spherical representations, and scenario-specific mapping functions. 2) *Scale- and shift-invariant loss function* decoupling depth learning from absolute scale by aligning predictions to ground truth through invariant optimization objectives.

These approaches aim to ensure mapping fidelity and robustness under diverse imaging conditions.

2.2.4 Ensuring Temporal Consistency: Temporal Stabilization with Sequential Reference

Motivation: In monocular videos, dynamic objects and camera motion can cause temporal flicker and inconsistencies between frames, degrading 3D reconstruction and visual stability.
Key Strategies: 1) *Temporal reasoning via architectural revision* integrating temporal-interaction layers, sliding-window refinement, or global video-context processing to preserve temporal coherence. 2) *Temporal-consistency constraint loss function* supervising temporal gradients, enforcing optical-flow-guided pixel alignment, or maintaining prediction deviations across frames.

These techniques improve temporal stability while balancing computational overhead.

3 Expanding Training Horizons: Multi-Source Data Extension

3.1 Pseudo-Label Generation in Semi-Supervised Learning

3.1.1 Basic Label Generation Paradigm

Pseudo-labeling, an effective semi-supervised learning technique, leverages unlabeled data for data expansion. The

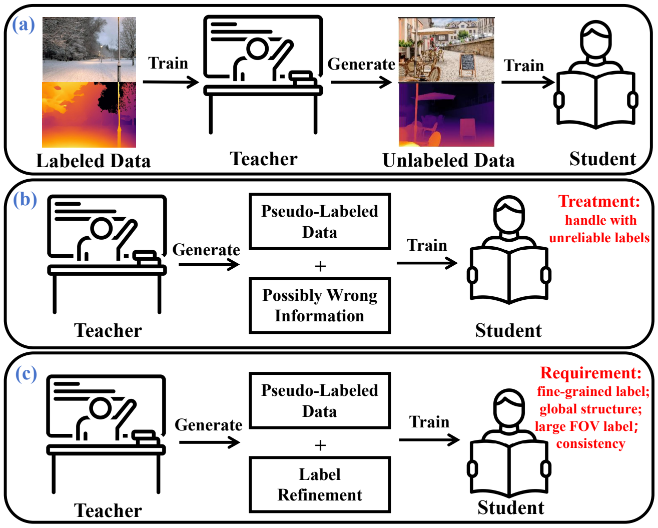

Fig. 2. Illustration for pseudo-label generation. (a) Basic paradigm for pseudo-label generation in semi-supervised learning; (b) Extended paradigm: discriminate unreliable regions in the pseudo labels and generate corresponding marks; (c) Extended paradigm: refine the generated labels to meet particular requirements.

standard pipeline (Fig. 2 (a)) has three stages: (1) a teacher model is trained on labeled data; (2) the teacher generates pseudo labels for unlabeled images; (3) a student model is trained with the pseudo-labeled data, resulting in superior generalization.

Depth Anything [18] exemplifies this paradigm. It enhances the teacher with the encoder from DINOv2 [30] and generates pseudo labels for 62M unlabeled images from eight diverse public datasets. A similar approach using 70-hour YouTube videos as the data source is presented in [31].

3.1.2 Extended Label Generation Paradigm

Pseudo-label filtering: To discriminate unreliable pseudo labels, studies [32], [33] design filtering mechanisms (Fig. 2 (b)). EndoOmni [32] uses a per-pixel pseudo-label confidence estimation, enabling the student to assign higher learning weights to more reliable labels. ER-Depth [33] proposes two consistency-based filters: a depth consistency filter to select pseudo labels robust against various perturbations, and a geometric consistency filter to select accurate and reliable pseudo labels.

Pseudo-label Refinement: Studies [20], [34]–[36] propose novel pseudo-label refinement strategies (Fig. 2 (c)) to meet the application demands. Distill Any Depth [34] integrates global and local depth cues for pseudo-label enhancement. The teacher processes both full images and multi-scale cropped patches, aided by a diffusion model [37]. FiffDepth [20] uses DINOv2 to generate pseudo labels and a learnable filter to solely preserve low-frequency regions, thereby guiding the student optimization for global structure learning. Depth Anywhere [35] generates pseudo labels for unlabeled 360-degree images via cube projection and masks invalid regions like sky and watermarks. StableDepth [36] normalizes pseudo labels to a unified scale and shift across the entire video to achieve scene consistency.

Summary: Pseudo-label generation expands the scale and diversity of training data, enabling foundation models to learn robust depth estimation from raw images and videos, thereby reducing reliance on manual annotations. However, this approach depends on a high-precision teacher model. Training such a model is time-consuming, and its errors can propagate and be amplified in the student model.

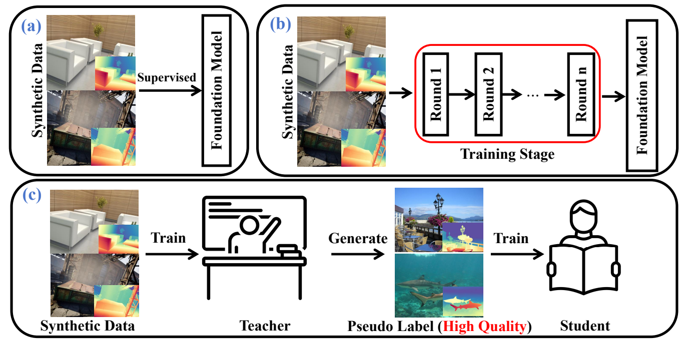

Fig. 3. Illustration of synthetic data usage. (a) Directly using synthetic data for model training; (b) Multi-round training with synthetic data integration: selectively incorporate synthetic data for training quality; (c) Teacher model enhancement for pseudo-label generation: enhance the quality of pseudo labels with synthetic data usage.

3.2 Leveraging Synthetic Data with Domain Adaptation

Real-world data is often limited by inaccurate labels from collection issues and coarse annotations [19]. In contrast, synthetic images are generated with highly precise depth labels [37]. They capture fine details and obtain accurate labels of challenging elements like transparent or reflective surfaces.

Given these advantages, several studies [38]–[40] rely exclusively on synthetic data for model training (Fig. 3) (a)). However, the noticeable gap still exists between synthetic and real images, particularly in terms of styles and color distributions, hindering effective transfer to real-world environments. To bridge this synthetic-to-real gap, various domain adaptation strategies have been developed.

3.2.1 Multi-round Training with Synthetic Data Integration

Several studies [41]–[43] employ multi-round training that selectively incorporates synthetic data (Fig. 3 (b)). For example, DepthPro [41] employs a two-stage curriculum: first training on a mixture of real labeled data to learn robust features; then retraining on synthetic data to enhance prediction with its pixel-perfect labels.

3.2.2 Teacher Model Enhancement via Synthetic Data Supervision

A common semi-supervision strategy, depicted in Fig. 3 (c), uses a teacher-student framework for synthetic-to-real adaptation [19], [44]–[46]. A teacher model trained on synthetic data generates better pseudo labels for unlabeled real images, which then supervise a student model. This allows the student to adapt to the real domain with higher estimation accuracy, as exemplified in Depth Anything V2 [19].

MoGe-2 [47] refines this approach, generating pseudo-depth values for real labeled images. Regions in the ground truth with significant mismatches against the pseudo-labels are filtered out and reconstructed using logarithmic-space

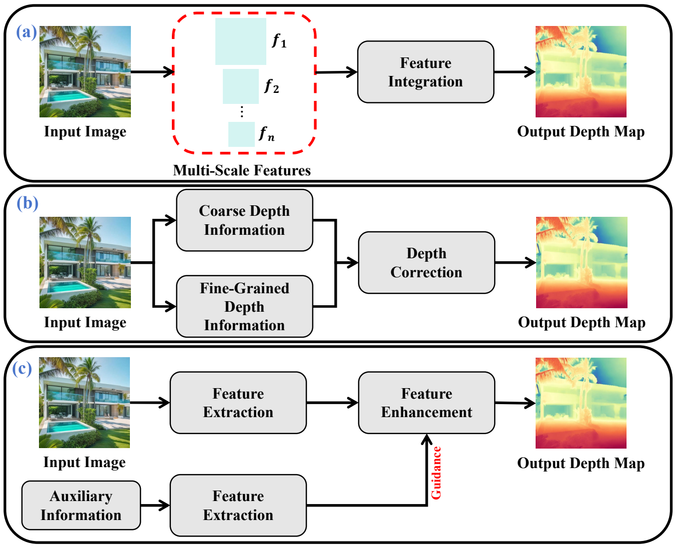

Fig. 4. Illustration of detail enhancement strategies. (a) Multi-scale feature fusion: extraction of image features across scales to preserve details and global structure; (b) Fine-grained information for detail correction: generation of fine-grained depth for original error correction; (c) Auxiliary-guided depth feature enhancement: depth feature refinement using auxiliary sources (e.g., text, segmentation and calibration).

Poisson completion. This process produces refined ground-truth depth maps with dense details.

Summary: Leveraging precisely annotated synthetic data, foundation models reduce depth prediction errors while capturing local details. Moreover, synthetic data enhances depth accuracy for transparent and reflective surfaces with specialized high-quality labels. Through adaptation strategies, the domain gap is bridged.

However, synthetic images exhibit limited scene coverage, as they are sampled from predefined, fixed scenes (e.g., living rooms and street scenes). Additionally, effectively integrating synthetic data in multi-round training requires dynamic adjustment based on multifaceted factors, such as the optimal ratio of heterogeneous data sources and mitigating the counteractive effects between real and synthetic data [19]. For the pseudo-labeling strategy, the core challenge remains the advent of high-capacity teacher models capable of producing reliable labels.

4 Preserving Geometric Fidelity: Detail Enhancement for Fine-Grained Depth

4.1 Local Structure Sharpening via Architectural Revision

Architectural innovations that enable foundation models to generate fine-grained geometric details (e.g., sharp edges) can be categorized into three paradigms. The first category [41], [48], [49] employs a multi-scale feature integration, which preserves local details across spatial contexts, as illustrated in Fig. 4 (a). Second, many approaches [40], [45], [50]–[57] introduce an additional branch to generate fine-grained depth, thereby correcting original unreliable regions, as given in Fig. 4 (b). Furthermore, some methods take guidance from auxiliary information sources to enhance spatial features [21], [43], [58]–[62], as visualized in Fig. 4 (c).

4.1.1 Multi-Scale Feature Fusion

Depth pro [41] employs a multi-scale ViT to integrate hierarchical depth features. The input image is down-sampled and divided into patches. These patches are processed by a shared encoder and the resulted features are transformed, up-sampled, and fused through a Dense Prediction Transformer (DPT) decoder [10] to produce the final depth. Meanwhile, Primedepth [49] enhances its fusion module with cross- and self-attention to refine multi-level features. In a different approach, DAR [48] formulates multi-scale feature integration as the auto-regression process, generating the depth map from low to high resolution to progressively improve structural coherence and detail.

4.1.2 Fine-Grained Information Generation for Detail Correction

There are mainly three strategies for fine-grained depth information generation, including: generation from 1) local image patches, 2) diffusion model process, and 3) frequency-domain transformation.

Local Image Patch: This strategy crops local patches for fine-grained information capturing. To refine an initial estimation, PatchRefiner [45] processes high-resolution patches to extract features, which are fused with coarse global features via convolution. Then, the fused representation is decoded to generate a residual correction map. Similarly, PatchFusion [50] introduces a Fusion Network that explicitly combines a coarse depth estimation with fine-grained depth from cropped patches, guided by multi-scale intermediate features. In addition, PRO [40] employs a patch-wise refinement with wavelet transforms [63] to integrate global structure and local details.

In a different approach for local patch generation, Multi-Depth [51] leverages various sampling strategies, including pixel-unshuffling, random subsampling, and segmentation masking. A shallow U-Net then processes these samples to yield individual refined depth maps, and the final output is synthesized by aggregating all sample predictions with the original estimation.

Diffusion Model Process: Leveraging the capacity for detail generation, diffusion models are integrated as an auxiliary branch to synthesize fine-grained information. SharpDepth [53] leverages a Noise-aware Gating mechanism to derive a difference map between the two-branch estimations, which guides depth fusion via the exponential moving average strategy. In addition, Perfecting Depth [52] employs an adaptive Sparsity Depth Refinement framework [64] to enhance sparse depth using the posterior estimates from the diffusion model. Furthermore, Better-depth [56] adopts the Stable Diffusion [65] to generate fine-grained depth for value refinement, guided by the original coarse estimation. Finally, GeometryCrafter [57] designs an extra residual encoder to capture geometric information lost during the original compressing process, aiding the diffusion model in fine-grained depth generation.

Frequency-Domain Transformation: Frequency-domain transformation techniques can effectively accentuate high-frequency details. DepthMaster [55] presents a Fourier Enhancement module that operates in the frequency domain via a 2D Fast Fourier Transform. This module adaptively refines low-frequency structural features with high-frequency

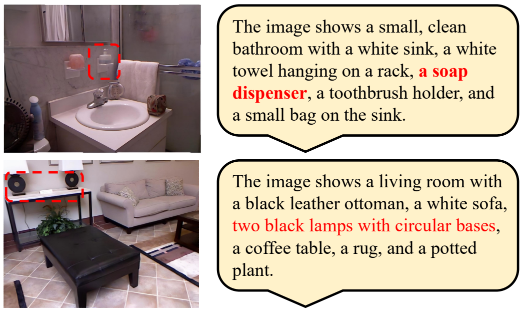

Fig. 5. Instance of images and corresponding description. The emphasis is marked red.

detail features. Similarly, the Laplacian Visual Prompting (LVP) module [54] amplifies edge features using a Laplacian operator and integrates predictions from the original and transformed counterparts.

4.1.3 Auxiliary-Guided Depth Feature Enhancement

Three primary sources provide auxiliary guidance for depth feature enhancement: image segmentation, text description, and camera calibration.

Image Segmentation: Several studies [21], [58], [61] leverage semantic information to guide the enhancement of depth features. BriGeS [58] employs a bridging gate, composed of cross- and self-attention blocks, to facilitate interaction between depth and semantic feature. Similarly, Amodal Depth Anything [21] incorporates a guidance convolution layer in parallel with the image feature convolution, allowing the incorporation of semantic masks, which convey complete object shape information. Finally, Pixel-Perfect Depth [61] injects the semantic representations into the Diffusion Transformer (DiT) blocks.

Text Description: PriorDiffusion [59] extracts language priors from a text encoder, covering object geometry, spatial relationships, and existence. These priors direct the model's attention to specific regions. Example images with corresponding textual descriptions are provided in Fig. 5. In addition, TPDepth [62] leverages text prompts to provide crucial semantic priors, using a controlled fusion mechanism for depth enhancement.

Camera Calibration: Camera information, particularly intrinsic parameters, is employed to enhance depth features [43], [60], helping produce geometry-aware features that improve spatial reasoning and emphasize physical shape.

Summary: Various architectural innovations address a key limitation of monocular depth estimation: over-smoothed predictions that lack details.

Multi-Scale Feature Fusion: This method effectively merges broad scene geometry from low-resolution features with fine details from high-resolution ones. It demonstrates strong architectural cohesion through a unified trainable network. However, it is computationally expensive due to the processing and fusion of large feature maps. Additionally, without careful design, features from different scales may introduce redundancy.

Fine-Grained Information for Detail Correction: This method explicitly recovers lost object boundaries and local structures. Nevertheless, it risks generating artifacts if the detailed information misaligns with the original prediction. Furthermore, employing secondary diffusion models or processing multiple high-resolution patches is computationally intensive and slows inference.

Auxiliary-Guided Depth Feature Enhancement: This modular approach enriches depth features by integrating auxiliary data (e.g., segmentation or camera calibration) via attention mechanisms or feature fusion. Its main drawback is the dependency on these auxiliary sources, which may not be available in purely monocular settings, limiting practical application.

4.2 Detail Preservation Loss Function

Detail preservation losses explicitly enforce the focus on fine details during model optimization and fall into three categories. First, edge-aware loss functions penalize prediction errors at high-gradient edges [20], [66]. Second, local region-based loss functions are computed within patches or spherical spaces to retain fine details [44], [57], [67], [68]. Finally, pixel pair-based loss functions maintain original depth relationships of labels by focusing on discrepancies between pixel pairs [45], [69].

4.2.1 Edge-Aware Loss Function

Gradient loss [20], [66] penalizes the discrepancy between predicted and ground truth values in high-gradient regions, defined as:

$$\mathcal{L}_{\text{grad}} = |\frac{\partial(D-\hat{D})}{\partial x}| + |\frac{\partial(D-\hat{D})}{\partial y}|, \tag{3}$$

where D and $\hat{D}$ are predicted and labeled depth maps.

In addition, Unidepthv2 [70] computes an edge-aware loss on salient edges, identified by the magnitude of RGB spatial gradients, defined as:

$$\mathcal{L}_{\text{EG-SSI}} = \sum_{\omega\in\Omega} ||\mathcal{N}_{\omega}(D_{\omega}) - \mathcal{N}_{\omega}(\hat{D}_{\omega})||_1, \tag{4}$$

where Ω is the boundary region, $D\omega$ and $\hat{D}\omega$ are the predicted and ground-truth depth maps within ω, and $\mathcal{N}_{\omega}$ denotes normalization by subtracting the median and dividing by the mean absolute deviation over ω.

4.2.2 Local Region-based Loss Function

The works [44], [67] introduce losses computed on cropped regions to direct the concentration of the training stage on local details. PanDA [44] designs a patch normalization loss, which is formulated as:

$$\mathcal{L}_{\text{EPNL}} = \frac{1}{N}\frac{1}{K}\sum_{i=1}^{N}\sum_{j=1}^{K}|\mathcal{N}_j(D_i) - \mathcal{N}_j(\hat{D}_i)|, \tag{5}$$

where N is the number of sampled images, K is the number of cropped regions per image, and $\mathcal{N}_j$ denotes depth normalization within the j-th region.

Depth Anything-AC [67] proposes Spatial Distance Relation (SDR) loss $\mathcal{L}_{\text{SDR}}$, which uses spatial distances between local patches to enforce inter-local variation. It is defined as:

$$\mathcal{L}_{\text{SDR}} = \frac{1}{N}\sum_{i=1}^{N}(S_D(D_i) - S_D(\hat{D}_i))^2, \tag{6}$$

where N denotes the image quantity, and S_D is the total local patch distance of the depth map, composed of position relation S_p and depth relation S_d:

$$S_D = \sqrt{S_p^2 + S_d^2}, \tag{7}$$

where S_p calculates the Euclidean distance between pixel coordinates of patch pairs, and S_d is defined as the absolute difference in predicted depth maps between patch pairs.

For foundation models, which estimate 3D point maps and calculate depth based on the coordinates, a local region-based loss quantifies positional deviations between aligned predicted and ground-truth point clouds within a spherical space V_j at the anchor point p_j [57], [68]. This space is defined as:

$$V_j = \{i \mid \|p_i - p_j\| \leq r_j, i \in \mathcal{M}\}, \tag{8}$$

$$r_j = \alpha \cdot z_j \cdot \frac{\sqrt{W^2 + H^2}}{2 \cdot f}, \tag{9}$$

where p_i are neighboring 3D points and V_j is defined with radius r_j. The radius is scaled relative to the depth z_j of p_j, focal length f, and image size $W \times H$ by scale factor α. In experiments, α uses three scales: $1/4, 1/16, 1/64$.

4.2.3 Pixel Pair-based Loss Function

Loss functions based on pixel-pair discrepancies are proposed in [45], [69]. PatchRefiner [45] employs a ranking loss to ensure predicted depth relationships match the labels:

$$\mathcal{L}_{rank} = \frac{1}{N} \sum_i \phi(p_{i,1}, p_{i,2}) \tag{10}$$

$$\phi(p_{i,1}, p_{i,2}) = \begin{cases} \log(1 + \exp(-\ell \times (d_{i,1} - d_{i,2}))), & \ell \neq 0, \\ (d_{i,1} - d_{i,2})^2, & \ell = 0, \end{cases} \tag{11}$$

where $(p_{i,1}, p_{i,2})$ is the i-th sampled point pair with predicted depths $(d_{i,1}, d_{i,2})$ and ground-truth depths $(\hat{d}i, 1, \hat{d}i, 2)$, N is the total number of sampled pairs, and ℓ is the ordinal class derived from the ground truth:

$$\ell = \begin{cases} +1, & \hat{d}_{i,1}/\hat{d}_{i,2} >= 1 + \tau, \\ -1, & \hat{d}_{i,1}/\hat{d}_{i,2} <= \frac{1}{1+\tau}, \\ 0, & \text{otherwise}, \end{cases} \tag{12}$$

where τ is a tolerance threshold, which is set to 0.03.

In addition, a sparse ordinal loss is presented in [69]. The formulation is given below:

$$\mathcal{L}_o(i,j) = \begin{cases} (\Delta d_{ij})^2 & \text{if} |\Delta \hat{d}_{ij}| < \delta, \\ \text{ReLU}(-\Delta d_{ij} \times \text{sgn}(\Delta \hat{d}_{ij})) & \text{otherwise}, \end{cases} \tag{13}$$

where $\Delta d_{ij} = d_i - d_j$, $\Delta \hat{d}_{ij} = \hat{d}_i - \hat{d}_j$, (d_i, d_j) and $(\hat{d}_i, \hat{d}_j)$ are predicted and ground-truth depths of a pixel pair (i, j), and δ is a small constant that prevents over-penalization of near-equal depths, sgn is the sign function.

Summary: These loss functions enhance depth map details by explicitly directing the model's attention to specialized regions and pixel-level relationships.

Edge-Aware Loss Function: This function effectively enhances the sharpness of boundaries. Its straightforward computational formulation allows for efficient integration into existing training pipelines. However, it is sensitive to label noise, which can be amplified during gradient calculation, propagating incorrect supervision signals and leading to inaccurate learning of edge characteristics.

Local Region-based Loss Function: This function directs the model's learning focus toward local parts and demonstrates significant adaptability with diverse local sampling strategies. However, it harms inter-regional consistency. Excessive concentration may introduce discontinuities at the junctions between adjacent areas.

Pixel Pair-based Loss Function: This function aims to preserve relative depth relationships between pixels, thereby maintaining depth variations. However, its computational complexity is a significant concern, as the number of pixel pairs considered increases quadratically with the image size. Consequently, comprehensive consideration of pixel sampling strategies is crucial to balance computational efficiency with functional performance.

5 Resolving Scale Ambiguity: Learning for Depth Mapping Fidelity

5.1 Geometric Revision in Pixel-Depth Mapping

To establish a clearer pixel-to-depth mapping under varying conditions, methods employ one of five core strategies. First, depth range narrowing directly normalizes label values to mitigate the domain gap [18], [38], [41], [43], [71], [72] (Fig. 6 (a)). Second, the canonical camera model transforms data into a unified space with a predefined focal length [17], [73], [74] (Fig. 6 (b)). Third, projection transformation simulates the inherent distortion of large-FOV cameras [44], [75], [76] (Fig. 6 (c)). Fourth, spherical representation separates camera intrinsics and spatial geometry [70], [77], [78] (Fig. 6 (d)). Finally, separate mapping functions are designed for specific sub-scenarios to handle scale ambiguity [39], [79] (Fig. 6 (e)).

5.1.1 Depth Value Range Narrowing

Studies [18], [41] use inverse depth to normalize values, defined as $d^* = 1/d$, which confines the range to $[0, 1]$. Orchid [71] further introduces a normalization step as $d' = d^*/d_\sigma$, where d_σ is the mean absolute deviation from the median. The values are then zero-shifted: $d_{\text{final}} = d' - d'_{\text{min}}$.

Works [38] employs a min-max scaling of the depth values across the dataset:

$$d' = \frac{d - d_2}{d_{98} - d_2}, \tag{14}$$

where d_2 and d_{98} are the 2% and 98% percentiles of depth values.

several previous works [43], [72] leverage log function for normalization. The calculation in [43] is defined as:

$$d_{\log} = \log_b \left((b-1) \frac{d - d_{\min}}{d_{\max} - d_{\min}} + 1 \right), \tag{15}$$

where b is the logarithm base. The log-scale normalization in [72] is defined as:

$$d_{\log} = \text{normalize} \left(\frac{\log(d/d_{\min})}{\log(d_{\max}/d_{\min})} \right), \tag{16}$$

$$\text{normalize}(d) = \text{clip}(2d - 1, -1, 1). \tag{17}$$

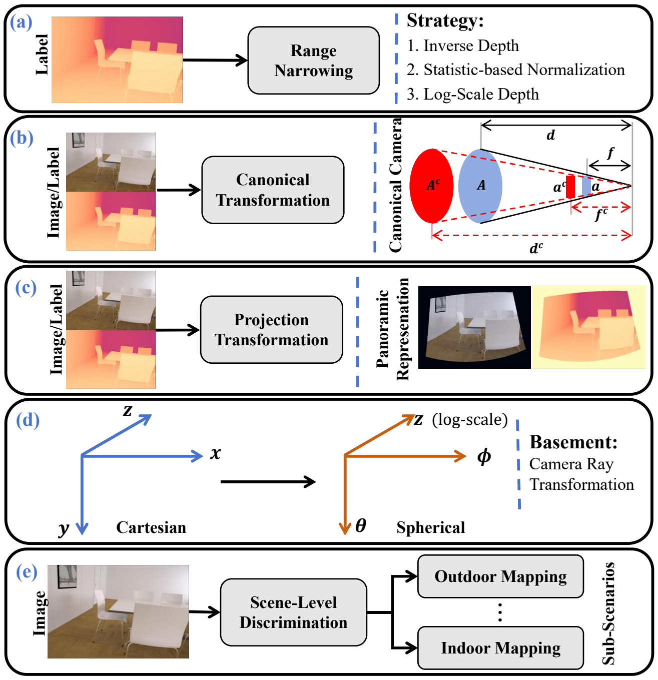

Fig. 6. Illustration of revision in pixel-depth mapping. (a) Depth value range narrowing; (b) Canonical camera model with unified focal length; (c) Projection transformation for distortion simulation; (d) Spherical representation for geometry reconstruction; (e)Separate mapping functions for sub-scenarios.

5.1.2 Canonical Camera Model with Unified Focal Length

The canonical camera model resolves depth ambiguity caused by varying camera focal lengths [17], [73], [74]. When images have similar appearances but different depths due to focal length variations, models struggle to learn a consistent pixel-depth mapping. Methods [17], [73] transform all training data into a canonical space with a predefined focal length f^c, standardizing the data as if captured by the same camera via label and image transformations.

1) Label transformation: For a real camera with focal length f, a ratio $\omega_d = \frac{f^c}{f}$ is defined. During training, the ground-truth depth $\hat{D}$ is scaled, as $\hat{D}_c = \omega_d \hat{D}$. The depth map is also resized, as $\hat{D}_c = \mathfrak{T}(\hat{D}, \omega_d)$, where $\mathfrak{T}(\cdot)$ denotes image resizing. In inference, the predicted depth D_c is transformed back to the original space with the ratio $\frac{1}{\omega_d}$.
2) Input image transformation: The images are transformed to simulate the canonical camera imaging. Specifically, image I is resized with the ratio $\omega_i = \frac{f^c}{f}$, as $I_c = \mathfrak{T}(I, \omega_i)$.

GVDepth [74] extends the canonical camera framework by introducing planar geometry priors to create a vertical canonical representation, where a target's depth is directly determined by its vertical position in the image.

5.1.3 Projection Transformation for Distortion Simulation

To bridge the gap between images of various FOVs, studies [44], [75] employ Equi-Rectangular Projection (ERP) to generate panoramic representations for standard perspective images. This transformation is based on Gnomonic Projection transformation [80], as depicted in Fig. 7.

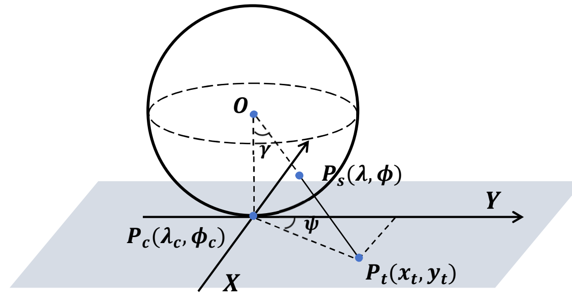

Fig. 7. Illustration of ERP conversion.

A point $P_s(\lambda, \phi)$ on the sphere (ERP space) centered at O is projected to a point $P_t(x_t, y_t)$ on a plane, which is tangent to the point $P_c(\lambda_c, \phi_c)$. Here, (λ, ϕ) are longitude and latitude, and (x_t, y_t) represents Cartesian coordinates on the tangent plane. For a tangent image with $\alpha \times \beta$ FOV and a side length of ω, the pixel coordinate (u_t, v_t) for $P_s(\lambda, \phi)$ is formulated as:

$$\begin{cases} u_t = \frac{w}{2} \cdot \left(\frac{x_t}{\frac{\beta}{360} \cdot \pi} + 1 \right) \\ v_t = \frac{w}{2} \cdot \left(\frac{y_t}{\frac{\alpha}{180} \cdot \frac{\pi}{2}} + 1 \right) \end{cases}, \tag{18}$$

where

$$\begin{cases} x_t = \frac{\cos\phi \sin(\lambda - \lambda_c)}{\cos\gamma} \\ y_t = \frac{\sin\phi\cos\phi_c - \cos\phi\sin\phi_c\cos(\lambda-\lambda_c)}{\cos\gamma} \\ \cos\gamma = \sin\phi\sin\phi_c + \cos\phi\cos\phi_c\cos(\lambda - \lambda_c) \end{cases}, \tag{19}$$

where, γ is the included angle between OP_c and OP_t.

These functions establish a one-to-one correspondence between ERP coordinate (λ, ϕ) and image coordinate (u_t, v_t). During training, both images and their labels are converted into the ERP space.

5.1.4 Spherical Representation for Geometry Reconstruction

To ensure foundation models learn a clear mapping function for 3D geometry, studies [70], [76], [77] introduce a pseudo-spherical representation that decouples camera intrinsics from spatial geometry.

The basic representation is defined by spherical coordinates (θ, ϕ, z), where θ and ϕ are the camera ray's azimuth and elevation, and z is logarithmically adjusted, calculated as follows:

$$(r_1, r_2, r_3) = K^{-1}[u, v, 1]^T, \tag{20}$$

$$(r_x, r_y, r_z) = Normalize(r_1, r_2, r_3), \tag{21}$$

$$\theta = \arctan(r_x/r_z), \phi = \arcsin(r_y), \tag{22}$$

where (u, v) is the image coordinate, K is the calibration matrix, and $Normalize$ denotes normalization process. Similarly, UniK3D [78] replaces z with Euclidean radius r, the distance between targets and the camera center. This ensures that object imaging size varies consistently and unambiguously with radial distance.

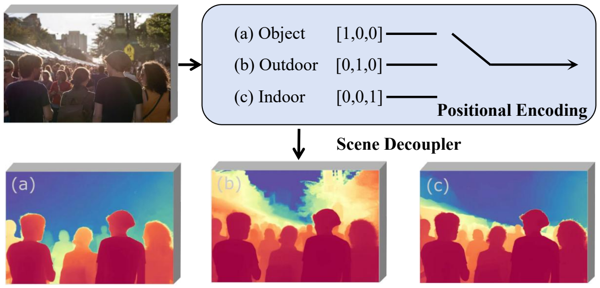

Fig. 8. The decoupler of GeoWizard. The decoupler designs one-hot vectors to achieve discrimination between various scenarios: object without background, outdoor scene, and indoor scene.

5.1.5 Separate Mapping Functions for Sub-Scenarios

Many methods [39], [79] address ambiguities by enabling models to discern different scenarios and learn a specialized mapping function for each.

GeoWizard [39] introduces a scene distribution decoupler that uses one-hot vectors to guide the model in learning distinct mapping functions for three scenario types: indoor, outdoor, and background-free objects (Fig. 8).

Similarly, Metric-Solver [79] employs a dual-decoder network. One decoder generates a normalized near-field depth map, while the other produces a tapered far-field map using an exponential normalization function with a negative exponent.

Summary: This overview summarizes strategies to address inconsistencies caused by varying camera parameters.

Depth Value Range Narrowing: This strategy serves as the most direct and widely adopted data processing method, which generates a more concentrated distribution to accelerate model convergence. However, the compressed values are relative and require post-processing to restore the true scale.

Canonical Camera Model with Unified Focal Length: This revision mitigates depth ambiguity caused by different focal lengths while preserving the metric scale. However, it necessitates precise camera calibration parameters, which limit the model's universality during inference.

Projection Transformation for Distortion Simulation: This strategy is tailored for large-FOV cameras, enabling multi-source data training. However, its computational complexity remains high due to intricate projection transformations that involve non-linear coordinate mapping.

Spherical Representation for Geometry Reconstruction: This method decouples the camera parameters and geometry information, creating two independent sub-tasks: depth estimation and camera ray estimation. This separation enables specialized optimization. However, it adds complexity to training loss, requiring separate supervision for the angle prediction.

Separate Mapping Functions for Sub-Scenarios: This strategy enables separate learning of mapping functions, mitigating the negative impact of heterogeneous depth distributions. However, it employs an overly simplistic classification that neglects intra-scenario variations. Moreover, the approach requires additional scenario labels to discriminate between different contexts.

5.2 Scale- and Shift-Invariant Loss Function

Scale- and shift-invariant losses are designed to set scene-agnostic learning objectives [18], [41], [67], [81], defined as:

$$\mathcal{L}_{ssi} = \frac{1}{M}\sum_{i=1}^{M}\rho\left(d_i^* - \hat{d}_i^*\right), \tag{23}$$

where ρ defines the specific loss functions, (e.g., mean absolute error), M denotes the pixel quantity of one image, and d_i^* and $\hat{d}_i^*$ are pixel-level depth of aligned predicted depth map $D^* \in \mathbb{R}^M$ and $\hat{D}^* \in \mathbb{R}^M$, respectively.

Let s and t be the scale and shift parameters. To compute the invariant loss, the prediction and ground truth must be aligned, which is achieved through two strategies.

1) Align the prediction to the ground truth based on the least-squares criterion.

$$(s,t) = \arg\min_{s,t}\sum_{i=1}^{M}\left(sd_i + t - \hat{d}_i\right)^2, \tag{24}$$

$$d_i^* = sd_i + t, \quad \hat{d}_i^* = \hat{d}_i, \tag{25}$$

where d_i and $\hat{d}_i$ are the original depth values of prediction and ground truth, respectively. s and t can be determined in closed form by rewriting (24) and (25) as a standard least-squares problem. Let $\vec{d_i} = (d_i, 1)^\top$ and $h = (s,t)^\top$, then the objective can be rewritten as:

$$h^{opt} = \arg\min_{\mathbf{h}}\sum_{i=1}^{M}\left(\vec{d_i}^\top h - \hat{d}_i\right)^2, \tag{26}$$

which has the closed-form solution:

$$h^{opt} = \left(\sum_{i=1}^{M}\vec{d_i}\vec{d_i}^\top\right)^{-1}\left(\sum_{i=1}^{M}\vec{d_i}\hat{d}_i\right). \tag{27}$$

2) Align the prediction with the ground truth by removing their translation and scaling them to unit size:

$$d_i^* = \frac{d_i - t(D)}{s(D)}, \quad \hat{d}_i^* = \frac{\hat{d}_i - t(\hat{D})}{s(\hat{D})}, \tag{28}$$

where $t(D) = \mathrm{median}(D)$, $s(D) = \frac{1}{M}\sum_{i=1}^{M}|d_i - t(D)|$.

Similarly, scale- and shift-invariant losses [47], [68] for point map estimation are defined as:

$$\mathcal{L}_{ssi-point} = \frac{1}{M}\sum_{i=1}^{M}\|p_i^* - \hat{p}_i^*\|_1, \tag{29}$$

$$p_i^* = sp_i + t, \quad \hat{p}_i^* = \hat{p}_i, \tag{30}$$

where p_i^* and $\hat{p}_i^*$ are pixel-level location of aligned predicted point map $P^* \in \mathbb{R}^M$ and $\hat{P}^* \in \mathbb{R}^M$, respectively.

Studies [41], [81] propose scale-invariant gradient loss, which is formulated as:

$$\mathcal{L}_{grad} = \sum_{k=1}^{K}\sum_{i=1}^{M}\left(|\frac{\partial(R_i^k)}{\partial x}| + |\frac{\partial(R_i^k)}{\partial y}|\right), \tag{31}$$

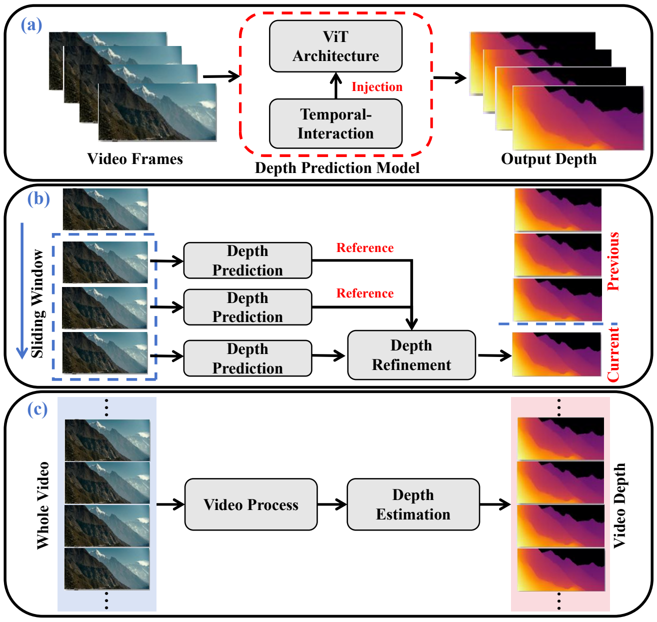

Fig. 9. Illustration of network revision for temporal stabilization. (a) Integration of temporal-interaction component; (b) Sliding-window framework for depth refinement; (c) Video-level context processing for global coherence.

where $R_i = d_i^* - \hat{d}_i^*$ is the depth difference, and R^k denotes the difference map at scale k, computed across $K = 4$ scales with resolution halved at each scale.

Summary: These loss functions address scale-dependent biases from varying camera parameters, allowing models to learn robust depth mappings that generalize across scenarios. However, a key drawback is the output of relative depths, which requires additional post-processing to recover the metric scale, thereby adding implementation complexity.

6 Ensuring Temporal Consistency: Temporal Stabilization with Sequential Reference

6.1 Temporal Reasoning via Architectural Revision

To mitigate flickering in monocular video depth estimation, three primary network revision strategies are used. First, the temporal-interaction component is incorporated to improve temporal modeling [82]–[84], as presented in Fig 9 (a). Second, many studies leverage a sliding-window framework to stabilize the current frame's depth using sequential information from neighboring frames within the window [85]–[89], as visualized in Fig. 9 (b). Finally, video-level context is processed to align depth predictions across the entire video [90]–[93], as illustrated in Fig. 9 (c).

6.1.1 Integration of Temporal-Interaction Component

Studies [82], [83] integrate temporal layers into depth prediction networks to enable temporal interactions.

Video Depth Anything [82] inserts temporal layers (comprising self-attention and feed-forward modules) into the DPT head. The generated head samples multi-scale features and fuses them from low to high resolution through the Fusion layer. The resulting fused maps are processed by the output layer to produce the final results. Similarly, Buffer Anytime [83] injects temporal layers into the feature fusion stage. Each temporal layer comprises several temporal attention blocks followed by a projection layer.

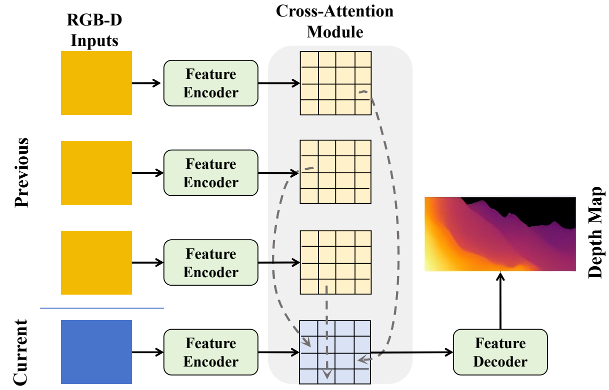

Fig. 10. Overview of the stabilization network. RGB-D Inputs are the combination of Video frames and corresponding depth maps.

In contrast, FlashDepth [84] adds a recurrent Mamba network after the DPT decoder to align features temporally, as defined by:

$$f, H_t = \mathrm{Mamba}(\mathrm{Flatten}(\mathrm{Down}(F)), H_{t-1}), \tag{32}$$

$$F_{\mathrm{align}} = F + \mathrm{Up}(\mathrm{UnFlatten}(f)), \tag{33}$$

where F is the DPT decoder's output feature of shape (B, C, H, W) and H_t is Mamba's hidden state at time t, propagated through the video, $\mathrm{Flatten}$ and $\mathrm{UnFlatten}$ operations reshape the tensor between (B, C, H, W) and $(B, H \times W, C)$, while Down and Up denote bilinear downsampling and upsampling, respectively.

6.1.2 Sliding-Window Framework for Depth Refinement

Several studies [85], [86] adopt a sliding-window framework to stabilize depth estimation, as shown in Fig. 10. A cross-attention module is employed to refine depth-aware features, based on temporal information from relevant frames. In this module, the current frame's features are queries and previous frames provide keys and values. The refined features are then passed to a decoder that generates consistent depth.

In addition, other methods focus on post-processing initial predictions. CH3Depth [87] employs a lightweight UNet to aggregate predictions within the window, while Depth Any Video [88] uses an interpolation network to refine intermediate frames, based on precise key frames.

Building upon Stable Video Diffusion [94], ChronoDepth [89] initializes overlapping frames of sliding windows with previously predicted depth and adapts the conditioning noise for each frame within the window. Consequently, consistent information is propagated across clips.

6.1.3 Video-Level Context Processing for Global Coherence

RollingDepth [90] scans the video to create key snippets with varying frame spacing. It aligns the depth predictions from these snippets by minimizing the differences between their shared frames.

StereoDiff [91] enforces global consistency through stereo matching. It pairs each frame with the subsequent n frames, resulting in a total of $nT - (n+1)n/2$ image pairs, where T denotes the total frame count. Each pair is processed through a stereo matching pipeline to generate coarse point maps, ensuring strong temporal consistency.

By adopting a 4D Geometry DiT, Sora3R [92] processes all video frames simultaneously to capture global spatio-temporal dependencies, thereby generating temporally consistent 3D geometry information. Additionally, UniGeo [93] concatenates the input video frames and the noisy geometry sequence into a single token sequence. This design allows the self-attention mechanism in the Diffusion Transformer structure [95] to process all frames and conditions jointly.

Summary: Various architectural strategies have been proposed to resolve the temporal flickering in video depth estimation.

Integration of Temporal-Interaction Component: This strategy integrates temporal layers or recurrent modules, enabling end-to-end training. It effectively achieves short-term consistency by modeling dependencies between adjacent frames. However, the approach has a limited receptive field and poor global coherence. The temporal receptive field of attention mechanisms or recurrent networks is typically constrained by computational costs. Furthermore, its reliance on neighboring frames makes it susceptible to localized errors.

Sliding-Window Framework for Depth Refinement: This strategy uses a sliding window that stabilizes depth estimation by using data from previous frames. It offers high flexibility, as the adjustable window size allows for a balance between computational cost and performance. A key limitation, however, is its sensitivity to motion blur: rapid movement of objects or the camera can cause significant disparities between reference frames and the current frame, leading to artifacts.

Video-Level Context Processing for Global Coherence: This strategy processes the entire sequence to achieve global consistency. While this enables optimization from a holistic perspective and minimizes error accumulation, it comes with high computational complexity. The massive memory and processing requirements make it unsuitable for long videos or real-time applications.

6.2 Temporal Consistency Constraint Loss Function

To stabilize flickering artifacts, specialized loss functions impose temporal constraints during training. These fall into three categories. First, the temporal gradient supervision loss matches the temporal gradients between prediction and ground truth [82]. Second, the optical flow-based stabilization loss enforces temporal depth consistency between aligned inter-frame pixels [83], [85], [86]. Finally, the temporal deviation maintaining loss preserves the prediction variance over time [87].

6.2.1 *Temporal Gradient Supervision Loss Function*

To enhance temporal consistency, Video Depth Anything [82] directly aligns the temporal gradients of predicted and ground-truth depth via a Temporal Gradient Matching (TGM) loss. This loss constrains the inter-frame difference in predicted depth to match the ground truth, formulated as:

$$\mathcal{L}_{\mathrm{TGM}} = \frac{1}{L-1}\sum_{i=1}^{L-1} \| \mid D^*_{i+1} - D^*_i \mid - \mid \hat{D}^*_{i+1} - \hat{D}^*_i \mid \|_1, \tag{34}$$

where L indicates the video length and $\| \cdot \|_1$ denotes $\ell 1$ distance. D^* and $\hat{D}^*$ are aligned depth prediction and ground truth, respectively.

6.2.2 *Optical Flow-Based Stabilization Loss Function*

To enforce temporal consistency, several methods use optical flow to align pixels across frames, ensuring depth predictions agree for corresponding pairs. Buffer Anytime [83] uses the GMFlow estimator [96] to calculate correspondence and defines a stabilization loss as:

$$\begin{aligned}\mathcal{L}_{\mathrm{stable}} = &\frac{1}{M}\sum_x |D_i(x) - D_{i+1}(O_{i\Rightarrow i+1}(x)+x)|_1 \\ &+ \frac{1}{M}\sum_x |D_{i+1}(x) - D_i(O_{i+1\Rightarrow i}(x)+x)|_1\end{aligned}, \tag{35}$$

where M denotes the pixel number per image and x indicates the image pixel. D_i and D_{i+1} are consecutive depth maps, while $O_{i\Rightarrow i+1}$ and $O_{i+1\Rightarrow i}$ are predicted optical flow maps, which indicate the adjacent pixel-level changes.

Studies [85], [86] adopt the optical flow-based warping loss [97] to supervise temporal consistency:

$$\mathcal{L}_t(i+1,i) = \frac{1}{M}\sum_x V_{i+1\Rightarrow i}(x)\|D_{i+1}(x) - D'_i(x)\|_1, \tag{36}$$

where D'_i is the predicted inverse depth D_i warped by the backward optical flow between input frames I_{i+1} and I_i. In the implementation, GMFlow [96] is adopted for optical flow estimation. $V_{i+1\Rightarrow i}$ is the visibility mask [97], calculated from the warping discrepancy between frame I_{i+1} and warped frame I'_i:

$$V_{i+1\Rightarrow i} = \exp(-\beta \|I_{i+1} - I'_i\|_2^2), \tag{37}$$

where β is set 50.

6.2.3 *Temporal Deviation Maintaining Loss Function*

Building on deviations between past predictions and ground truth within a sliding window, a temporally consistent deviation loss is proposed [87], defined as:

$$\mathcal{L}_{tmp} = |\mathcal{L}_0 - \frac{1}{w}\sum_{i=0}^{w}\mathcal{L}_i|, \tag{38}$$

$$\mathcal{L}_i = \mathbf{MSE}(D^*_{k-i}, \hat{D}^*_{k-i}), \tag{39}$$

where w denotes the sliding window size, D^*_{k-i} and D^*_{k-i} are aligned predicted and ground-truth depth maps at frame $k-i$, respectively, and $\mathbf{MSE}$ calculates the mean square errors. This loss ensures the stability of prediction errors over time.

Summary: These loss functions aim to ensure stable and coherent depth prediction across video sequences.

Temporal Gradient Supervision Loss: This function aligns the temporal gradients of predicted and ground-truth depth. While computationally efficient and easy to integrate, it is sensitive to errors in the ground truth and struggles with rapid motion or scene changes due to its reliance on frame-by-frame differences.

Optical Flow-Based Stabilization Loss: This function employs optical flow to enforce depth consistency at corresponding pixels across frames, making it effective for dynamic scenes. A significant limitation is its reliance on the accuracy of the precomputed optical flow, since errors within the flow fields directly propagate to the depth estimation.

Temporal Deviation Maintaining Loss: By measuring the prediction deviations across a sliding window, this function ensures stability over multiple frames. While it successfully mitigates frequent flickering, it may also cause the persistent impact of initial depth errors on subsequent predictions.

7 Adaptation Strategies for Foundation Models

Despite progress in general-purpose depth foundation models, two challenges hinder real-world deployment. First, trained primarily on common imagery, these models often underperform in challenging environments like endoscopy or low-light scenes. Second, trained with the relative depth objective, these models struggle to infer accurate metric distance, limiting their practical utility. Therefore, adaptation is essential.

In this article, we review related adaptation strategies, which can be classified into three categories. First, many studies directly adjust the pre-trained foundation models with target-domain data [66], [98]–[102], as demonstrated in Fig 11 (a); Second, lightweight trainable component is injected to networks for adaptation [103]–[107], as illustrated in Fig 11 (b). Finally, scaling parameters are calculated with auxiliary information to transform depth value [108]–[115], as visualized in Fig. 11 (c).

7.1 Direct Fine-Tuning with Target-Domain Data

In studies [98], [99], the EndoSLAM [116] dataset is used to directly adjust the pre-trained model, enhancing its performance in medical domains. Other approaches [100], [101] incorporate uncertainty estimation, adding branches that generate confidence maps to guide the fine-tuning process.

Additionally, DepthDark [102] retrains the foundation model for low-light conditions [117] using illumination guidance. First, a grayscale version of low-light images is generated to provide a stable luminance signal. The original low-light images and illumination guidance are then combined and processed to aid the fine-tuning stage.

Finally, to adapt the pre-trained foundation model [19] for metric estimation, Prompt Depth Anything [66] employs sparse LiDAR as the metric prompt to guide the retraining process. This is implemented through a multi-scale prompt fusion architecture that integrates LiDAR prompts into the DPT decoder through zero-initialized convolutional layers.

7.2 Integration of Lightweight Trainable Component

Several studies [103]–[107] employ lightweight, trainable components for parameter-efficient fine-tuning while keeping the base foundation model frozen. This strategy effectively adapts foundation models to specific niche domains.

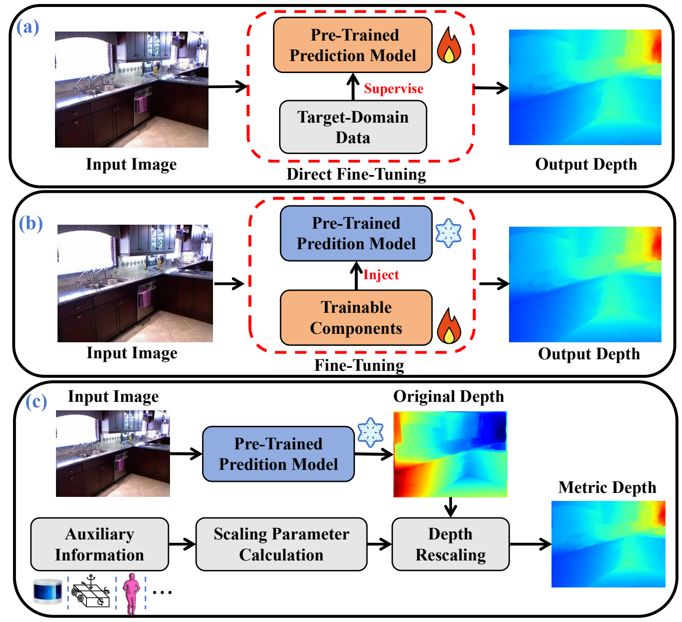

Fig. 11. Illustration of foundation model adaptation strategies. (a) Direct fine-tuning with target-domain data; (b) Integration of lightweight trainable component; (c) Scaling parameter generation with auxiliary guidance, e.g., sparse measurements.

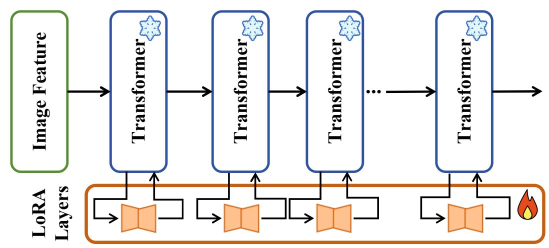

Fig. 12. Illustration of fine-tuning with LoRA.

A common approach is Low-Rank Adaptation (LoRA) [118], which is commonly used to fine-tune foundation models [103]–[105], [107]. It integrates trainable low-rank matrices into Transformer layers to significantly reduce the number of parameters requiring updates [103]–[105], as illustrated in Fig. 12. Additionally, the study [106] appends lightweight, trainable ”calibration tokens” to each encoder layer. This aligns the distribution of fisheye image embeddings with that of normal images, adapting the pre-trained model for fisheye cameras.

7.3 Scaling Parameter Generation with Auxiliary Guidance

By incorporating the scaling priors of specific scenarios, the output of the foundation model can be effectively translated into metric results. Existing methods leverage various prior sources.

Works [108], [109] leverage sparse measurements from sensors (e.g., low-resolution LiDAR) or from SFM with poses provided by an IMU. RANSAC algorithm [119] is employed to estimate scaling and shifting parameters by fitting the 3D sparse measurements to the depth estimation [108].

These parameters are subsequently used to transfer the original prediction. Furthermore, the work [109] segments the scene (via SAM [120]) and computes per-region scaling and shifting factors.

Studies [110], [111] leverage large language models to generate scale priors from scene descriptions. Rsa [110] uses the CLIP text encoder [121] to process language inputs, while TR2M [111] introduces a cross-modality attention module to generate scale information. In this module, image feature embeddings (queries) retrieve scale information from text feature embeddings (keys/values).

To obtain metric scale priors, the human mesh recovery [122] is adopted to estimate human dimensions from processed images [112], thereby propagating scale from human pixels to the full scene, achieving transformation for relative depth predictions. TanDepth [113] adopts data from the Global Digital Elevation Model (GDEM) to provide absolute-scale priors for UAV perception. HybridDepth [114] leverages a depth-from-focus (DFF) method [123] to provide crucial metric information for scale prior calculation. Following this similar pipeline, defocus blur is adopted to generate metric depth cues [115].

Summary: This section outlines strategies for adapting pre-trained foundation models for real-world applications.

Direct Fine-Tuning with Target-Domain Data: This standard approach is straightforward to implement and applicable to all model adjustments. However, it is highly data-dependent, requiring extensive annotated data, and is computationally intensive as the entire model is updated.

Integration of Lightweight Trainable Component: This approach is parameter-efficient and computationally inexpensive, as it freezes the base model and only trains small, added modules. This makes it ideal for resource-constrained environments while preserving the pre-trained model's robust feature extraction capabilities. However, a key limitation is its architectural dependency, as it requires a compatible model design.

Scaling Parameter Generation with Auxiliary Guidance: This strategy uses external data to generate scaling parameters, enabling metric outputs without retraining. While practical, it is highly dependent on external data, limiting its applicability. Furthermore, inaccuracies in the auxiliary information directly degrade the depth estimation.

TABLE 1
Comparison of zero-shot performance on the NYU-Depth V2 and KITTI datasets, where the top and the bottom of the table list results of relative and metric depth estimation performance, respectively. † denotes methods of monocular video depth estimation.

Model	NYU-Depth V2		KITTI	
	ARel↓	δ_1↑	ARel↓	δ_1↑
Orchid [71]	5.7	96.9	7.7	94.4
DepthAnything-AC [67]	5.1	97.3	8.3	93.4
BetterDepth [56]	4.2	98.0	7.5	95.2
FiffDepth [20]	4.4	97.8	7.3	93.5
DepthMaster [55]	5.0	97.2	8.2	93.7
Lotus [124]	5.4	96.8	8.5	92.2
DepthAnything V2 [19]	4.4	97.9	7.5	94.8
Marigold [38]	5.5	96.4	9.9	91.6
GeoWizard [39]	5.2	96.6	9.7	92.1
PrimeDepth [49]	5.8	96.6	7.9	93.7
PriorDiffusion [59]	5.9	96.0	10.4	90.3
BriGes [58]	4.2	**98.2**	7.4	94.7
GenPercept [37]	5.9	96.7	7.8	93.5
Metric3D [17]	5.0	96.6	5.8	97.0
Metric3D V2 [73]	4.3	98.1	**4.4**	**98.2**
DepthFM [46]	5.5	96.3	8.9	91.3
DepthAnything [18]	4.3	98.1	7.6	94.7
Distill Any Depth [34]	4.3	98.1	7.0	94.9
Pixel-Perfect Depth [61]	**4.1**	97.7	7.0	95.5
TPDepth [62]	5.0	96.8	8.9	93.5
Monst3r† [125]	9.1	88.8	10.1	89.3
DepthAnyVideo† [88]	5.1	97.0	7.3	95.1
CH3Depth† [87]	5.0	97.2	7.6	94.9
VideoDA† [82]	4.6	97.8	7.5	94.6
ChronoDepth† [89]	7.3	94.1	-	-
DepthCrafter† [42]	7.2	94.8	-	-
GRIN [43]	5.8	98.0	**4.6**	**98.3**
SM4Depth [126]	12.6	96.0	8.7	92.8
SharpDepth [53]	6.0	96.9	6.0	97.3
ZoeDepth [117]	8.0	95.2	6.0	96.5
ZeroDepth [60]	10.0	90.1	6.4	95.8
Metric3D [17]	9.4	92.6	5.8	96.4
UniDepth [77]	**4.3**	**98.4**	4.7	98.0
Metric3D V2 [73]	6.1	98.0	5.1	97.7

8 EXPERIMENTAL ANALYSES

This section presents extensive quantitative comparisons of previous MDFMs across depth accuracy, detail preservation, anti-perturbation performance, and downstream task adaptation. To ensure fairness, we report only representative methods evaluated on identical benchmarks and experimental evaluating splits. The results are sourced from public benchmarks or original publications. For clarity in the tables, the best results are shown in bold and the second-best are underlined.

8.1 Evaluation for Depth Precision

Two primary types of metrics are used to evaluate performance:

- **Depth Accuracy:** Thresholded accuracy is defined by $max(\frac{d_i}{d_i^*}, \frac{d_i^*}{d_i}) = \delta < t$, where t is a given threshold and $t\epsilon(1.25, 1.25^2, 1.25^3)$.
- **ARel**: Absolute relative error is defined by $\frac{1}{|M|}\sum_{i\epsilon M}\frac{|d_i - d_i^*|}{d_i^*}$.

The primary evaluation datasets are listed below:

- Outdoor KITTI dataset [127] consists of data captured in rural, highway, and urban areas, featuring complex scenes with traffic elements.
- Indoor datasets, including NYU-Depth V2 [128], Scannet [129], and Bonn [130], contain scenes of living rooms and studying halls under varying lighting conditions.

8.1.1 *Evaluation of Monocular Images*

Table 1 presents the zero-shot depth estimation performance of representative MDFMs on NYU-Depth V2 and KITTI. As shown, for relative depth estimation, the top-6 methods are

TABLE 2
Comparison of zero-shot video depth estimation performance on the Scannet and Bonn datasets.

Model	Scannet		Bonn	
	ARel↓	δ_1↑	ARel↓	δ_1↑
Video Depth Anything [82]	**7.5**	**95.4**	5.3	97.2
Depth Any Video [88]	11.2	88.3	**5.1**	**98.1**
NVDS [85]	18.7	67.7	16.7	76.6
DepthCrafter [42]	12.3	85.6	6.6	97.9
ChronoDepth [89]	15.9	78.3	10.0	91.1
RollingDepth [90]	9.3	91.6	7.9	93.9
Buffer Anytime [83]	12.3	85.3	10.2	92.5
StableDepth [36]	9.4	92.1	6.3	97.9

TABLE 3
Comparison of zero-shot boundary accuracy on the Spring, Sintel, and iBims datasets.

Model	Sintel F1↑	Spring F1↑	iBims F1↑
ZoeDepth [117]	2.7	0.1	3.5
Metric3D V2 [73]	32.1	2.4	9.6
PatchFusion [50]	31.2	3.2	13.4
UniDepth [77]	31.6	-	3.9
DepthAnything [18]	26.1	4.5	12.7
DepthAnything V2 [19]	22.8	5.6	11.1
Marigold [38]	6.8	3.2	14.9
Depth Pro [41]	40.9	7.9	17.6
FiffDepth [20]	**42.3**	**8.6**	**18.9**

TABLE 4
Comparison of the robustness of foundation models against various perturbations. The reported metrics represent the average performance indicators calculated across all disturbed scenarios.

Model	Average Error↓	Accuracy Stability↓
MiDaS [81]	2.25	0.40
Depth Anything [18]	1.88	0.32
Depth Anything V2 [19]	1.69	0.29
ZoeDepth [117]	2.42	0.41
UniDepth V2 [70]	1.40	**0.27**
Metric3D V2 [73]	1.70	0.29
Marigold [38]	2.17	0.40
Depth Pro [41]	**1.24**	0.30
MoGe [68]	1.46	0.31

BriGes [58], BetterDepth [56], Metric3D V2 [73], DepthAnything [18], Distill Any Depth [34], Pixel-Perfect Depth [61] on NYU-v2, and Metric3D V2 [73], Metric3D [17], Depth Any Video [88], Distill Any Depth [34], BetterDepth [56], Pixel-Perfect Depth [61] on KITTI, respectively.

Besides, we can conclude that: 1) The leading performance of DepthAnything and Distill Any Depth underscores the effectiveness of pseudo-label generation. By leveraging large-scale unlabeled data, foundation models obtain strong zero-shot capabilities. 2) Building on the framework of DepthAnything, both BriGes and BetterDepth achieve further improvement. BriGes enhances depth features by incorporating semantic information [120], whereas BetterDepth refines details using the Diffusion model. 3) Using the canonical camera model, the Metric3D series learns a robust pixel-wise depth mapping function that generalizes across various domains. 4) With the aid of semantic information, diffusion-based model exhibits competitive performance.

For zero-shot metric depth estimation on NYU-Depth V2 and KITTI, the leading models are UniDepth [77] and GRIN [43], both of which leverage camera parameters to guide depth generation. UniDepth predicts camera parameters directly from images to prompt depth estimation, while GRIN uses the given parameters to serve as the conditioning signal for the diffusion model.

8.1.2 Evaluation of Video Sequences

For monocular video depth estimation, following existing studies, we report the zero-shot performance of representative foundation models on Bonn [130] and Scannet [129] video datasets. The quantitative comparisons are given in Table 2. As observed, Video Depth Anything [82] exhibits the most competitive performance with the insertion of temporal layers and constraint of TGM loss. The followed foundation models are Depth Any Video [88], DepthCrafter [42], and StableDepth [36].

8.2 Accuracy in Detail Preservation

To evaluate detail preservation, the boundary accuracy F1-Score [41] is used. The boundary sharpness of predicted depth maps is evaluated on standard benchmarks: Spring [132] and Sintel [133] (synthetic, movie), and iBims [134] (real-world, indoor). Table 3 shows the zero-shot performance of representative foundation models. We observed that the top models are FiffDepth [20] and Depth Pro [41]. FiffDepth uses a refined Stable Diffusion model and pseudo-label refinement to enhance geometric information for fine-grained prediction. Depth Pro excels through multi-scale feature integration and a multi-round training process on both real and synthetic data.

8.3 Robustness against Perturbations

We also compare the robustness of MDFMs against various perturbations. Aligning with previous studies [135], we compare results with controlled perturbations (object, camera, material, lighting) on synthetic datasets generated by Infinigen Nature [136] and Indoor [137]. For a base scene and M perturbations, let D_0, $\hat{D}_0$ be base depth prediction and ground truth, and D_i, $\hat{D}_i$ (for $i = 1, ..., M$) be their perturbed counterparts, we report results using the following two metrics:

Average Error μ quantifies the mean depth prediction error across all scenes, including both the base scene and perturbed scenes, defined as: $\mu = \frac{1}{M+1}\sum_{i=0}^{M}\mathbf{ARel}(D_i, \hat{D}_i)$.

Accuracy Stability σ denotes the variance of the depth errors over the perturbations and the base scene, defined as: $\sigma = \frac{1}{M}\sum_{i=0}^{M}(\mathbf{ARel}(D_i, \hat{D}_i) - \mu)^2$.

The robustness performance of prior models is summarized in Table 4. In this benchmark, Depth Pro [41] achieves the lowest average error against the ground truth, while UniDepth V2 [70] demonstrates the best stability results. Among the scale-invariant models for which self-consistency applies, UniDepth V2 achieves the best score, followed closely by MoGe [68].

8.4 Results of Model Adaptation

To evaluate the adaptation approaches, we report the relative depth accuracy achieved when adapting the foun-

TABLE 5
Performance of adapting foundation models to the SCARED dataset [131], where Paras.(M) denotes the total trained parameters.

Model	ARel↓	δ_1↑	Paras.(M)
Zero-Shot [18]	8.4	93.0	-
Directly Fine-Tuning [18]	5.8	97.4	11.2
DVLoRA [107]	5.1	98.1	1.66
RVLoRA [103]	**4.8**	**98.2**	**1.38**

TABLE 6
Performance of model adaptation for metric estimation on the NYU-Depth V2 dataset.

Model	ARel↓	δ_1↑
Directly Fine-Tuning [18]	5.6	**98.4**
LiDAR (1 beam) [108]	6.3	93.9
LiDAR (16 beams) [108]	**3.9**	97.6
LiDAR (32 beams) [108]	4.0	97.4
Text Description [110]	14.7	78.6

dation model to a niche medical domain (Table 5). Second, we assess metric depth estimation performance on the NYU-Depth V2 dataset (Table 6) where the scaling parameters are generated from LiDAR [108] and text description [110]. Both settings use Depth Anything as the basic model.

We find that DVLoRA [107] and DVLoRA [103] incorporating lightweight, trainable components to the frozen foundation model effectively fine-tune models for niche domains, delivering stronger performance with limited data. Moreover, as demonstrated in Table 6, method [108] leverages scaling parameters from precise auxiliary sources, like LiDAR scans, to convert relative estimations to metric scale, achieving accuracy comparable to or superior to elaborate fine-tuning. This highlights the strong potential for deploying foundation models on LiDAR-equipped robotics platforms.

8.5 Performance of MDFMs in Downstream Tasks

We then investigate the performance of MDFMs in downstream tasks. Specifically, we follow BenchDepth [138] and report results of MDFMs in stereo matching, depth completion, feed-forward monocular 3D scene reconstruction, and SLAM.

Table 7 summarizes the performance rankings of some representative MDFMs across the four downstream tasks, from which we can know that 1) Most MDFMs improve the performance of downstream tasks, highlighting their potential for practical applications, 2) Depth Anything V2 [19] is the only method that consistently improves performance across all downstream tasks, demonstrating the benefits of scaling up training data and incorporating synthetic data, and 3) The performance improvement from GenPercept [37] underscores the importance of effective fine-tuning strategies for Stable Diffusion-based models.

9 Challenges and Future Directions

9.1 Data Diversity and Availability

The development of MDFMs depends on large-scale, high-quality datasets. Although the pseudo-label strategy and introduction of synthetic data are explored comprehensively, they remain limited by the imperfect quality of labels from pre-trained teacher models and a lack of variability in rare or complex scenarios. Future research should focus on advanced physics-based synthetic generation [139], efficient annotation pipelines [140], [141], and domain-aware self-supervision learning [142], thereby facilitating MDFMs to perform robustly across all operational domains.

9.2 Accuracy for Thin, Small, and Distant Objects

Although the strategies of preserving geometric details are well explored in MDFMs [45], [58], depth accuracy still degrades greatly for thin and small objects, which are common in UAV perception, surveillance, and medical analysis. To address these challenges, specialized attention mechanisms for thin and tiny objects, along with effective super-resolution techniques for information enhancement, are potential directions for future research.

9.3 Zero-Shot Metric Depth Estimation

Mainstream MDFMs, which belong to the relative depth estimation model, typically require fine-tuning process or domain-specific priors (e.g., scaling parameters) to produce metric depth outputs. Existing representative solutions either rely on additional calibration data to resolve scale ambiguity [17], [60], [73] or exhibit significant performance degradation in unseen scenarios or with different camera parameters [77], [117], [126].

These limitations hinder their scalability and ease of deployment in real-world applications. Beyond simply scaling and diversifying training datasets, future research should prioritize the development of zero-shot metric depth foundation models. Such models would be capable of autonomously estimating camera parameters or domain-specific scaling factors. Furthermore, with the advancement of vision-language models in spatial understanding, adapting these models for metric depth estimation using appropriate prompts presents a highly promising research direction [143].

9.4 Temporal Consistency with Viewpoint Shifts

In autonomous driving or robot navigation, significant viewpoint shifts caused by camera motion are common, posing challenges for sequential reference and temporal consistency [144]. With advances in camera pose estimation, research has shown that temporal consistency can be enhanced using this auxiliary information [145], [146]. Future work on MDFMs should focus on leveraging camera pose estimation to mitigate the negative impacts of dynamic viewpoint changes. Promising approaches include using an off-the-shelf estimator as an auxiliary branch or developing a joint pose and depth learning framework with mutual reinforcement.

9.5 Continual Learning

Most current MDFMs follow a batch learning paradigm, trained once on available datasets and deployed with fixed parameters. Given the limited scale and diversity of existing data relative to real-world domains, there is a critical need

TABLE 7
Comparison of foundation model performance on the four downstream proxy tasks. $imp.$ indicates the average improvement ratio expressed in percentage and $avg.\ rank$ denotes the average rank of performances in all downstream tasks.

Model	Depth Completion		Stereo Matching		Monocular 3DGS		SLAM		$avg.\ rank$
	$imp.$	$rank$	$imp.$	$rank$	$imp.$	$rank$	$imp.$	$rank$	
Midas [81]	+3.09	3	-3.07	6	**+5.24**	**1**	+2.32	5	3.75
Depth Anything V2 [19]	**+9.26**	**1**	**+5.77**	**1**	<u>+4.21</u>	<u>2</u>	**+10.00**	**1**	**1.25**
Metric3D V2 [73]	-0.38	7	-1.74	5	-0.05	3	-4.19	7	5.5
UniDepth [77]	+2.97	4	-3.68	7	-0.10	4	<u>+7.08</u>	<u>2</u>	4.25
Marigold [38]	+1.76	5	+1.87	4	-4.19	7	+4.67	4	5
GenPercept [37]	<u>+6.16</u>	<u>2</u>	+1.99	3	-0.14	5	+6.16	3	<u>3.25</u>
Moge [68]	+1.53	6	<u>+2.70</u>	<u>2</u>	-1.60	6	-4.04	6	5

for models that continuously adapt to new targets. Although some prior works [147], [148] have explored continual learning for MDE, none address MDFMs. Advancing continual learning techniques could enable incremental domain adaptation without catastrophic forgetting, enhancing their applicability in dynamic, real-world settings.

9.6 Multi-task Generalization

While multi-task learning with auxiliary tasks like semantic segmentation has improved depth estimation [39], [49], [71], [93], achieving effective integration remains a major challenge. A primary obstacle is the difficulty of unifying depth estimation with other complementary visual tasks within a single, cohesive foundation model. Thus, developing efficient strategies for cross-task knowledge integration is a critical direction for future work.

9.7 Inference with Limited Resources

MDFMs are computationally expensive due to their complex architectures and refinement modules, demanding substantial hardware resources. However, real-world applications such as autonomous driving, robot navigation, and UAV perception require real-time inference under strict computational constraints. While techniques including knowledge distillation [149], lightweight model substitution [150], diffusion model reformulation [124], and memory cache design [151] have been explored, achieving real-time performance on resource-limited robotic platforms without sacrificing accuracy of MDFMs remains an open challenge.

9.8 Accuracy for Complex Materials

Current MDFMs struggle to accurately estimate depth for reflective and transparent surfaces (e.g., glass and mirrors in vehicles) due to complex optical properties and limited visual cues. Although prior work [152] has employed binary masks as input for transparent object identification, robust segmentation remains difficult. Moreover, shadows and illumination variations across diverse materials and conditions further reduce accuracy. Future research may develop frameworks that effectively exploit contextual knowledge of material properties, lighting, and object semantics, enabling improved robustness and accuracy in complex visual environments.

9.9 Unified Model across various FOV Cameras

While MDFMs demonstrate promising generalization within standard pinhole camera setups, their accuracy often degrades when applied to heterogeneous cameras such as fisheye, panoramic, or ultra-wide-angle models [76]. Conversely, specialized foundation models, while achieving ideal performance in large-FOV scenarios, are suboptimal for standard cameras and exhibit significant performance drops [78]. Therefore, future work should focus on unifying depth estimation models across camera types. This could involve integrating distortion-aware architectures with lens distortion priors to ensure precise feature-geometry correspondence, as well as expanding training diversity through cross-FOV augmentation and synthetic data generation.

9.10 Unified and Comprehensive Benchmark

While evaluation metrics and datasets for depth accuracy are relatively mature, the field of MDFMs currently lacks widely accepted benchmarks for assessing critical performance aspects such as temporal consistency, detail preservation, operational stability, and practical utility. Furthermore, research on depth foundation models suffers from discrepancies in performance metrics and evaluation datasets. Inconsistencies in training data, model scale, and computational overhead further complicate fair and comprehensive comparisons between these models.

10 Conclusion

Monocular Depth Foundation Models (MDFMs) have rapidly advanced by exploiting large-scale data, powerful architectures, and cross-domain learning to achieve remarkable generalization. This survey offers the first focused synthesis of the field, introducing a taxonomy that connects core challenges to key innovations, reviewing adaptation strategies for diverse scenarios, and outlining evaluation protocols that move beyond accuracy alone. Despite significant progress, issues such as data diversity, temporal stability, and zero-shot metric prediction remain open, while opportunities lie in multi-task integration, efficient continual adaptation, unified benchmarking, etc. By capturing the current state and listing future directions of MDFMs, we aim to guide their evolution toward models that are accurate, robust, and ready for real-world deployment.

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